

Project Introduction

Project Title:

Data-Driven Testing of SauceDemo Login Functionality Using Selenium WebDriver, TestNG, and CSV File.

1. Introduction

Software testing is an important phase of the Software Development Life Cycle (SDLC) that helps ensure the quality, reliability, and functionality of an application before it is released to end users. Testing helps identify defects, validate requirements, and improve the overall user experience.

This project focuses on testing the Login functionality of the SauceDemo web application. The Login module is one of the most critical components of any web application because it controls user authentication and access to the system. If the login functionality fails, users will be unable to access application features, leading to business and user-impacting issues.

The project combines both Manual Testing and Automation Testing approaches. Manual Testing is used to understand requirements, create test scenarios, design test cases, execute exploratory tests, and validate application behavior. Automation Testing is implemented using Selenium WebDriver, TestNG, and Java to execute login tests automatically using multiple data sets stored in a CSV file.

The project follows a Data-Driven Testing approach where different username and password combinations are maintained externally in a CSV file. This allows testers to execute the same test script with multiple test data sets without changing the automation code, thereby improving maintainability, reusability, and test coverage.

2. Application Under Test (AUT)

Application Name: SauceDemo

Application Type: Web-Based E-Commerce Testing Application

Application URL:

<https://www.saucedemo.com>

SauceDemo is a sample e-commerce application widely used for learning and practicing software testing concepts. It provides a realistic environment for validating login

functionality, product management, cart operations, checkout processes, and other e-commerce workflows.

For this project, testing is limited to the Login Module.

3. Project Objective

The primary objective of this project is to validate the Login functionality of the SauceDemo application using both manual and automated testing techniques.

The project aims to:

- Verify that users can successfully log in with valid credentials.
 - Verify that invalid credentials are rejected appropriately.
 - Validate system behavior for empty and incorrect input values.
 - Ensure proper error messages are displayed for failed login attempts.
 - Execute login tests using multiple data combinations.
 - Implement Data-Driven Testing using CSV files.
 - Generate automated execution reports.
 - Capture screenshots for passed and failed test executions.
 - Improve testing efficiency through automation.
 - Demonstrate practical implementation of Selenium Automation Framework concepts.
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4. Project Scope

The scope of this project includes testing the Login functionality of the SauceDemo application.

The following activities are included within the scope:

Manual Testing Activities:

- Requirement Analysis
- Test Planning

- Test Scenario Creation
- Test Case Design
- Test Data Preparation
- Test Execution
- Defect Reporting
- Test Closure Activities

Automation Testing Activities:

- Selenium WebDriver Automation
- TestNG Integration
- Data-Driven Testing
- CSV Data Handling
- Page Object Model (POM)
- Screenshot Capture
- Extent Report Generation
- Maven Build Execution
- Automated Test Execution

The following activities are outside the scope of this project:

- Product Module Testing
- Shopping Cart Testing
- Checkout Functionality Testing
- Performance Testing
- Security Testing
- Mobile Application Testing

5. Testing Approach

This project follows a hybrid testing approach consisting of both Manual Testing and Automation Testing.

Manual Testing Approach:

Initially, requirements are analyzed to understand the expected behavior of the login functionality. Test scenarios and test cases are prepared based on business requirements. Various positive and negative test cases are executed manually to identify potential defects.

Automation Testing Approach:

After manual validation, automation scripts are developed using Selenium WebDriver and TestNG. Test data is stored in a CSV file and read dynamically during execution. The automation framework executes login tests for multiple credential combinations and records execution results automatically.

6. Technologies Used

The following technologies and tools are used in this project:

Programming Language:

Java

Automation Tool:

Selenium WebDriver

Testing Framework:

TestNG

Build Tool:

Maven

Development Environment:

Eclipse IDE

Data Source:

CSV File

Reporting Tool:

Extent Reports

Version Control:

Git and GitHub

Operating System:

Windows

Browser:
Google Chrome

Driver Management:
WebDriverManager

7. Data-Driven Testing Concept

Data-Driven Testing is a testing technique where test data is separated from the test script and stored in external files such as CSV, Excel, Database, or XML.

In this project, login credentials are stored in a CSV file. During execution, Selenium reads each row from the CSV file and executes the login test repeatedly using different username and password combinations.

Benefits of Data-Driven Testing:

- Increased test coverage
 - Reduced code duplication
 - Easy maintenance
 - Improved reusability
 - Faster execution
 - Better scalability
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8. Expected Outcomes

Upon successful completion of this project:

- Login functionality will be validated thoroughly.
- Test scenarios and test cases will be documented.
- Defects will be identified and reported.
- Selenium automation scripts will execute multiple test data combinations.
- CSV-based Data-Driven Testing will be implemented successfully.
- Screenshots will be captured automatically.

- Execution reports will be generated.
 - TestNG reports and Extent Reports will provide detailed execution summaries.
 - The project will demonstrate both Manual Testing and Automation Testing skills.
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9. Benefits of the Project

This project provides practical exposure to real-world software testing activities.

Benefits include:

- Understanding Software Testing Life Cycle (STLC)
- Requirement Analysis Experience
- Test Case Design Skills
- Defect Reporting Practice
- Selenium Automation Implementation
- Framework Development Knowledge
- Data-Driven Testing Experience
- Reporting and Documentation Skills
- Interview Preparation for QA and Automation Testing Roles

The project serves as a complete demonstration of end-to-end testing activities from requirement analysis to automated execution and reporting.